Enhancing the Hosting Capacity of Photovoltaic System Considering BESS for Distribution Network

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***Abstract*—The rapid integration of distributed rooftop solar photovoltaic (PV) systems presents significant operational challenges for low-voltage distribution networks, including localized overvoltages and thermal congestion. This paper proposes an enhanced multi-period optimization framework based on Linear Power Flow (LPF) integrated with Battery Energy Storage Systems (BESS) and machine learning-based day-ahead solar forecasting. Evaluated on a synthetic 26-bus low-voltage network in Da Nang, Vietnam, the proposed predictive storage dispatch approach eliminates traditional conservative worst-case assumptions, successfully unlocking end-of-line bottleneck nodes and achieving up to a 72.0% increase in gross PV hosting capacity compared to the baseline network without storage.**

***Index Terms*—Hosting Capacity, Photovoltaic Systems, Battery Energy Storage Systems (BESS), Linear Power Flow, Machine Learning, Optimal Power Flow.**

I. Introduction & State of the Art

The global transition toward renewable energy has driven a rapid expansion of distributed energy resources, particularly rooftop solar photovoltaic (PV) systems.

In Vietnam, this momentum is explicitly guided by the national Power Development Plan VIII (PDP8), which encourages self-consumption rooftop solar with capacity targets reaching 46,459 to 73,416 MW by 2030 and up to 295,646 MW by 2050. However, low-voltage distribution networks were originally engineered for unidirectional power flow, moving electricity from centralized transmission substations down to passive end-use consumers.

When local solar production exceeds consumption during sunny midday hours, excess power flows in reverse back toward the upstream grid. This sudden surplus causes severe operational hurdles for Distribution Network Operators, including local bus overvoltages at remote nodes, thermal overloading of lines and transformers, and power quality degradation.

To quantify the maximum amount of solar generation a grid can absorb without violating these technical constraints or requiring immediate physical infrastructure upgrades, engineers rely on the concept of Hosting Capacity. While several levers exist to enhance hosting capacity, such as grid reinforcement, inverter-based voltage regulation, and generation curtailment, Battery Energy Storage Systems (BESS) offer a particularly compelling alternative. By charging during peak production hours and discharging during evening demand peaks, BESS units smooth net load profiles without requiring costly infrastructure reinforcement.

In recent literature, hosting capacity assessment methods are broadly categorized into deterministic, stochastic, and optimization-based approaches [1], [2]. Deterministic methods rely on fixed worst-case assumptions that frequently yield overly conservative hosting capacity estimates. To account for real-world uncertainties, such as fluctuating consumer demand profiles and random PV placement, stochastic methods using Monte Carlo simulations have been widely adopted [1], [2].

Optimization-based methods model hosting capacity as an Optimal Power Flow problem to maximize renewable integration. To address the heavy computational burden associated with non-linear AC power flow equations, the Linear Power Flow (LPF) formulation [3] was introduced to linearize voltage drop relationships while maintaining high numerical precision. Building upon this approach, L. H. Le and N. K. Le [4] developed an optimization framework that evaluates hosting capacity using an efficient two-step LPF model, demonstrating that linear programming can rapidly determine exact capacity bounds across low-voltage distribution networks.

Despite its computational efficiency, Assoc. Prof. Le Hong Lam's original model operated as a single-period static formulation focused primarily on steady-state peak conditions. This was a serious limitation as static single-period models cannot capture the dynamic, multi-period interaction between temporal load profiles and solar generation over time [2], [4].

Furthermore, without storage assets to buffer local generation, static optimization often assigns zero PV capacity to vulnerable bottleneck buses, an allocation strategy that is unenforceable in real customer-driven installations [4]. Traditional dynamic models that attempt to resolve this issue also frequently operate under the assumption of perfect foresight regarding next-day solar irradiance profiles, failing to account

for real-time solar intermittency and cloud transients in day-ahead operations [1].

To address these limitations, our project directly builds upon and improves Assoc. Prof. Le Hong Lam's original model through a three-stage engineering enhancement centered on a distribution network case study in the city of Da Nang, Vietnam.

First, we upgraded the single-period LPF formulation into a full multi-period optimization framework evaluated across a 24-hour operational horizon. Second, we integrated dynamic BESS models into this multi-period GAMS formulation. By charging during peak production hours and discharging during evening demand peaks, BESS units smooth net load profiles without requiring costly infrastructure reinforcement.

Finally, we implemented a predictive Machine Learning pipeline in Python trained on historical Da Nang weather data. This neural network generates accurate day-ahead solar forecasts that feed directly into the multi-period GAMS optimization model, replacing unrealistically perfect solar assumptions with predictive operational intelligence to guide optimal storage dispatch.

II. Mathematical Model Formulation

To evaluate the dynamic hosting capacity of the low-voltage distribution grid, the single-period static Linear Power Flow (LPF) optimization model developed by L. H. Le and N. K. Le was extended into a multi-period dynamic formulation over a 24-hour horizon ($t \in T = \{1, 2, \ldots, 24\}$).

This section details the mathematical formulation of the Battery Energy Storage System (BESS) integration using the exact variable, parameter, and scalar definitions declared in the GAMS model (`LINEAR_PF_STAGE_2_WITH_HC_V2.gms`).

A. Multi-Period Active Power Balance Integration

In the baseline static model, active power flow was evaluated at a single steady-state snapshot. In the extended formulation, time-series vectors for hourly load scaling (Load_profile(t)) and machine-learning-predicted day-ahead solar generation (PV_profile(t)) are introduced.

For every bus f and time step t, BESS charging power ($x_{\mathrm{Pch}}(f,t)$) acts as active power consumption (energy absorption), while discharging power ($x_{\mathrm{Pdis}}(f,t)$) acts as active power injection. The active power balance equation P_PF_e(f,t) is expressed in per-unit (p.u.) as:

$$\begin{aligned}
&\sum_{g} x_{\mathrm{Pg}}(f,g,t) - \sum_{d} \frac{P_d(f,d)}{\text{baseMVA}} \cdot \text{Load_profile}(t) \\
&\quad + x_{\mathrm{PDG}}(f) \cdot \text{PV_profile}(t) - x_{\mathrm{Pch}}(f,t) + x_{\mathrm{Pdis}}(f,t) \\
&\quad - \sum_{\text{tbus}} \Big[\text{Ybus_re}(f,\text{tbus}) \cdot \big(V(f,t) - V(\text{tbus},t)\big) \\
&\qquad + \text{Ybus_re}(f,\text{tbus}) \cdot \text{del_V_li}(f) \cdot \big(V(f,t) - V(\text{tbus},t)\big) \\
&\qquad - \text{Ybus_im}(f,\text{tbus}) \cdot \big(\delta(f,t) - \delta(\text{tbus},t)\big) \Big] = 0
\end{aligned} \tag{1}$$

where:

- $x_{\mathrm{Pg}}(f,g,t)$ is generator power injection.
- $P_d(f,d)$ is the normalized baseline bus demand.
- $x_{\mathrm{PDG}}(f)$ is the installed PV capacity at bus f.
- $x_{\mathrm{Pch}}(f,t)$ and $x_{\mathrm{Pdis}}(f,t)$ are the BESS charge and discharge variables.
- del_V_li(f) is a linearization correction factor for voltage deviations at bus f.
- The final summation calculates net linearized power flow leaving node f using real and imaginary admittance matrices, voltage deviations, and phase angles.

B. BESS State of Charge (SoC) Temporal Continuity

The energy stored within a BESS unit at node bus and hour t, tracked by the positive variable $x_{\mathrm{Ebat}}(\mathrm{bus},t)$ (in MWh), depends on its energy state at $t-1$, adjusted for charging/discharging rates and conversion efficiency (eff_bat $= 0.95$).

The energy continuity equation is enforced for all non-slack buses bus $\notin$ ref:

$$x_{\mathrm{Ebat}}(\mathrm{bus},t) = \begin{cases} \text{Ebat_max}\,0.50 + \left(\text{eff_bat}\, x_{\mathrm{Pch}}(\mathrm{bus},t) - \dfrac{x_{\mathrm{Pdis}}(\mathrm{bus},t)}{\text{eff_bat}}\right)\text{baseMVA}, & t = 1, \\ x_{\mathrm{Ebat}}(\mathrm{bus},t-1) + \left(\text{eff_bat}\, x_{\mathrm{Pch}}(\mathrm{bus},t) - \dfrac{x_{\mathrm{Pdis}}(\mathrm{bus},t)}{\text{eff_bat}}\right)\text{baseMVA}, & t > 1. \end{cases} \tag{2}$$

where:

- eff_bat $= 0.95$ yields an effective AC round-trip efficiency of $\approx 90.25\%$.
- Multiplication by baseMVA converts power variables (p.u.) to MW.
- At $t = 1$, initial energy is set to 50% of Ebat_max.

C. BESS Operational Constraints

1) Power Rating Limits: Maximum continuous charging and discharging rates are capped by Pbat_max $= 0.05$ MW (50 kW):

$$0 \le x_{\mathrm{Pch}}(\mathrm{bus},t) \le \frac{\text{Pbat_max}}{\text{baseMVA}} \tag{3}$$

$$0 \le x_{\mathrm{Pdis}}(\mathrm{bus},t) \le \frac{\text{Pbat_max}}{\text{baseMVA}} \tag{4}$$

2) State of Charge Operating Window: Stored energy is bounded between $\mathrm{SoC_min} = 0.10$ (10%) and $\mathrm{SoC_max} = 0.90$ (90%):

$$\mathrm{SoC_min}\cdot\mathrm{Ebat_max} \leq x_{\mathrm{Ebat}}(\mathrm{bus}, t) \leq \mathrm{SoC_max}\cdot\mathrm{Ebat_max} \tag{5}$$

The BESS operational parameters reflect standard industrial practice for Lithium-ion systems. The State of Charge operating window is bounded between 10% and 90% to protect cell health. The power rating of $50\,\mathrm{kW}$ and nominal capacity of $200\,\mathrm{kWh}$ establish a 4-hour continuous discharge duration, tailored to absorb excess solar generation throughout the peak midday irradiance window.

3) Nodal PV Capacity Limit: Installed PV capacity per bus is capped by $\mathrm{PDG_max} = 0.10$ MW (100 kW):

$$0 \leq x_{\mathrm{PDG}}(\mathrm{bus}) \leq \frac{\mathrm{PDG_max}}{\mathrm{baseMVA}} \quad \forall \mathrm{bus} \tag{6}$$

4) Slack Bus Exclusion: Battery deployment and PV allocation are zeroed at the reference node:

$$x_{\mathrm{PDG}}(\mathrm{ref}) = 0, \quad x_{\mathrm{Pch}}(\mathrm{ref}, t) = 0, \quad x_{\mathrm{Pdis}}(\mathrm{ref}, t) = 0 \tag{7}$$

D. Daily Cyclic Boundary Constraint

To guarantee sustainable daily operational cycles, final SoC at $t = 24$ must match initial SoC at $t = 1$:

$$x_{\mathrm{Ebat}}(\mathrm{bus}, 24) = 0.50 \cdot \mathrm{Ebat_max} \quad \forall \mathrm{bus} \notin \mathrm{ref} \tag{8}$$

E. Dynamic Objective Function Formulation

The objective maximizes total installed active solar capacity ($\mathrm{tot_PDG}$):

$$\max \mathrm{tot_PDG} = \sum_{\mathrm{bus}} \left(x_{\mathrm{PDG}}(\mathrm{bus}) \cdot \mathrm{baseMVA}\right) \tag{9}$$

subject to power flow balance, BESS continuity, branch capacity, and statutory voltage limits.

III. Results and Discussion

To evaluate the performance of the proposed multi-period optimization framework, numerical simulations were executed on a synthetic 26-bus low-voltage distribution network modeled after the grid infrastructure in Da Nang, Vietnam.

This section outlines the experimental setup, presents global network hosting capacity results across operating scenarios, analyzes nodal capacity distributions, and examines the dynamic State of Charge behavior of the installed battery units.

A. Data Specification and Experimental Setup

The optimization model integrates realistic historical meteorological datasets and residential load profiles to capture spatio-temporal power flow variations.

In an initial version of the model, the photovoltaic profile was obtained using the PVGIS Europe tool https://re.jrc.ec.europa.eu/pvg_tools/fr/. The location was set to Da Nang (Latitude: 16.06, Longitude: 108.212) and July was selected as a representative summer month. The resulting daily irradiance profile was then normalized and used as a static, representative PV_profile for the whole simulation.

In a later stage, this single-month approach was replaced by a machine learning-based day-ahead forecasting approach. Hourly solar irradiance records spanning 2005 to 2023 were extracted from PVGIS for Da Nang, corresponding to approximately 6,938 daily profiles.

A Multi-Layer Perceptron Regressor (MLPRegressor) with a single hidden layer of 60 neurons was trained using an 80% historical data split and validated against the remaining 20%, achieving a mean absolute error (MAE) of approximately 0.05 on the normalized profile.

Table I presents the photovoltaic production profile obtained in July in Da Nang.

TABLE I: Photovoltaic Production Profile (July, Da Nang)

t	Time (UTC+7)	$G(i)$	Normalized
t_1	00:30	0.00	0.00
t_2	01:30	0.00	0.00
t_3	02:30	0.00	0.00
t_4	03:30	0.00	0.00
t_5	04:30	0.00	0.00
t_6	05:30	7.27	0.01
t_7	06:30	99.12	0.13
t_8	07:30	263.08	0.35
t_9	08:30	436.13	0.59
t_{10}	09:30	586.31	0.79
t_{11}	10:30	691.20	0.93
t_{12}	11:30	744.81	1.00
t_{13}	12:30	733.03	0.98
t_{14}	13:30	661.52	0.89
t_{15}	14:30	528.09	0.71
t_{16}	15:30	363.50	0.49
t_{17}	16:30	202.85	0.27
t_{18}	17:30	61.78	0.08
t_{19}	18:30	0.00	0.00
t_{20}	19:30	0.00	0.00
t_{21}	20:30	0.00	0.00
t_{22}	21:30	0.00	0.00
t_{23}	22:30	0.00	0.00
t_{24}	23:30	0.00	0.00

Similarly, a normalized 24-hour demand profile (Load_profile) was implemented. Table II presents the normalized residential load profile of Vietnam.

B. Global Network Hosting Capacity Evaluation

Table III summarizes the gross installed solar capacity ($\mathrm{tot_PDG}$) achieved across each operating scenario solved

TABLE II: Normalized Residential Load Profile of Vietnam

t	Hours	Demand (MW)	Normalized
t_1	00:30	24 000	0.74
t_2	01:30	23 500	0.72
t_3	02:30	23 000	0.71
t_4	03:30	23 000	0.71
t_5	04:30	23 500	0.72
t_6	05:30	25 000	0.77
t_7	06:30	27 500	0.85
t_8	07:30	31 000	0.95
t_9	08:30	31 500	0.97
t_{10}	09:30	31 500	0.97
t_{11}	10:30	30 000	0.92
t_{12}	11:30	30 000	0.92
t_{13}	12:30	31 500	0.97
t_{14}	13:30	32 500	1.00
t_{15}	14:30	32 500	1.00
t_{16}	15:30	32 000	0.98
t_{17}	16:30	31 500	0.97
t_{18}	17:30	30 500	0.94
t_{19}	18:30	29 500	0.91
t_{20}	19:30	29 000	0.89
t_{21}	20:30	28 500	0.88
t_{22}	21:30	27 500	0.85
t_{23}	22:30	25 500	0.78
t_{24}	23:30	24 500	0.75

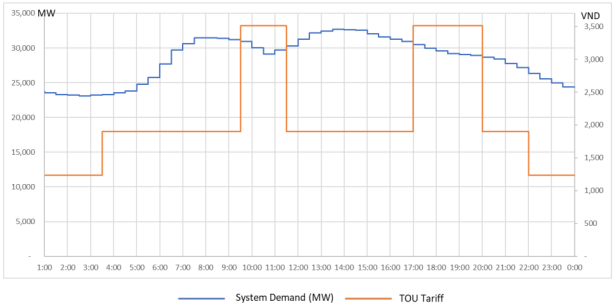


Figure 10. Viet Nam's Typical Load Profile (30' interval) vs TOU tariffs

Fig. 1: Load profile of Vietnam.

using the CPLEX linear programming solver in GAMS.

TABLE III: System-Wide PV Hosting Capacity Comparison

Scenario	**Gross PV (tot_PDG)**	**Gain**
Scenario 1 (No BESS)	1.25 MW	Baseline
Scenario 2 (BESS, No ML)	1.98 MW	+58.4%
Scenario 3a (ML, July)	2.11 MW	+68.8%
Scenario 3b (ML, Full Year)	2.15 MW	+72.0%

In the static baseline scenario without battery storage (Scenario 1), rigid steady-state power flow constraints cap the total allowable solar installation at 1.25 MW.

Midday solar peaks induce severe bus overvoltage along downstream feeders, forcing generation curtailment to prevent exceeding the upper voltage statutory threshold. Transitioning to a multi-period framework and deploying dynamic BESS flexibility without machine learning inputs (Scenario 2) increases allowable solar penetration to 1.98 MW, representing a 58.4% capacity gain over the baseline.

Integrating predictive day-ahead machine learning forecasts further enhances system performance. When trained on July meteorological data (Scenario 3a), the model achieves an installed PV capacity of 2.11 MW, marking a 68.8% increase.

Expanding the training pipeline to encompass the full multi-year historical dataset (Scenario 3b) yields the optimal outcome of 2.15 MW, representing a total 72.0% increase over the baseline.

This predictive management strategy provides an 8.6% improvement over standard dynamic storage control.

C. Nodal Capacity Allocation and Feeder Bottleneck Resolution

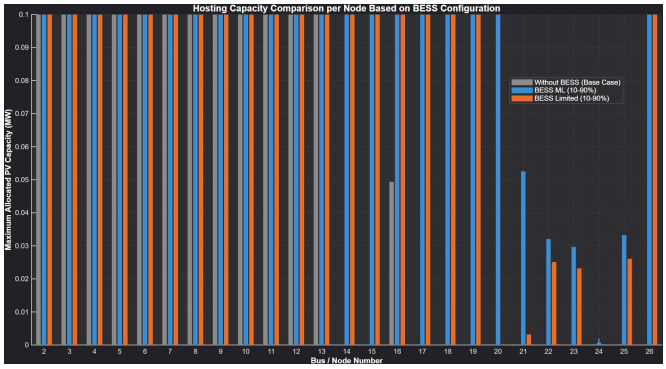


Fig. 2: Nodal hosting capacity allocation across the 26-bus network.

A bus-by-bus spatial analysis across the 26-node distribution network highlights how predictive storage dispatch resolves severe localized feeder bottlenecks.

Under static operation (Scenario 1) and standard dynamic BESS control (Scenario 2), remote feeder nodes 17, 18, and 20 represent critical network dead zones. High cumulative line impedance between these distant buses and the substation causes local voltage levels to rise rapidly under solar injection.

Because standard reactive storage control algorithms charge batteries prematurely during early morning hours, they lack sufficient remaining capacity during midday peaks. Consequently, conventional control strategies assign 0 MW of PV capacity to nodes 17, 18, and 20 to maintain voltage compliance.

In contrast, the predictive ML control framework (Scenario 3b) successfully unlocks these end-of-line dead zones, allocating the maximum permissible PV capacity of 0.10 MW to nodes 17, 18, and 20.

By anticipating peak midday solar irradiance, the predictive Energy Management System maintains necessary storage volume prior to solar peak hours. Across all 26 network

buses, the predictive machine learning approach matches or exceeds the allocated capacity of standard control strategies, demonstrating superior thermal congestion prevention and network-wide voltage stabilization.

D. Dynamic BESS Operating Mechanics

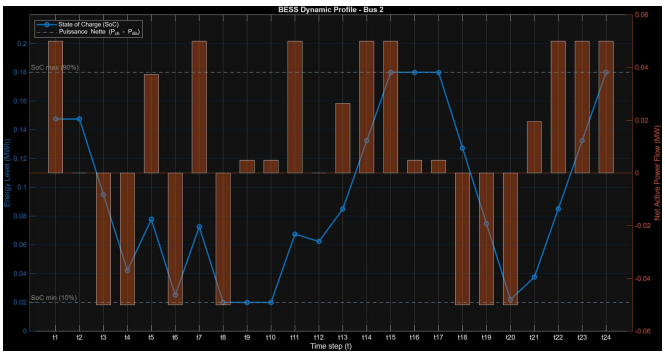


Fig. 3: 24-hour dynamic BESS operational profile at Bus 2.

Examining the time-series State of Charge and net power flow profiles at Bus 2 illustrates the physical operational mechanism of predictive storage dispatch. The optimal daily operation unfolds across three distinct phases.

During the early morning hours (t_8 to t_{10}), conventional BESS controllers frequently absorb energy, saturating storage capacity prematurely. The predictive ML forecast identifies the magnitude of the upcoming solar peak and holds the battery State of Charge at its minimum operational bound ($\mathrm{SoC_min} = 10\%$, or 0.02 MWh) through hour t_{10}. This pre-emptive reservation guarantees maximum available storage volume for midday absorption.

Throughout the maximum solar irradiance window (t_{11} to t_{15}, from 11:00 to 15:00), the BESS engages in intensive charging at full rated power ($\mathrm{Pbat_max} = 50$ kW). The stored energy climbs rapidly from 0.02 MWh to its upper limit ($\mathrm{SoC_max} = 90\%$, or 0.18 MWh). Localized energy absorption suppresses reverse power flow, keeping nodal voltages below 1.05 p.u. without requiring generation curtailment.

Finally, stored energy is held through late afternoon and discharged back into the distribution grid during the evening residential load peak (t_{18} to t_{20}, from 18:00 to 20:00). Discharging mitigates net feeder demand, supports local consumption, and satisfies the daily cyclic boundary condition, returning the State of Charge to exactly 50% (0.10 MWh) at hour t_{24} to ensure sustainable continuous daily operation.

E. Predictive Control vs. Worst-Case Baselines

It is essential to emphasize that integrating predictive machine learning does not inherently expand physical network hosting capacity under all theoretical conditions.

Instead, the significant performance gains observed in this framework arise because conventional non-predictive evaluations (Scenario 2) rely on deterministic, worst-case irradiance and load assumptions to guarantee grid compliance. These pessimistic static bounds force standard control logic to behave reactively, charging batteries during early morning hours to mitigate conservative risk thresholds.

Consequently, storage units saturate prematurely, exhausting available capacity before the critical midday solar peak occurs and triggering unnecessary generation curtailment.

Driving the multi-period optimization with day-ahead solar forecasts generated by the Multi-Layer Perceptron model replaces rigid worst-case profiles with realistic, time-varying generation trajectories. The CPLEX solver leverages this predictive visibility to optimize BESS scheduling proactively, holding battery State of Charge at its lower bound during morning hours and concentrating storage absorption strictly during peak irradiance.

Rather than altering physical grid limits, the machine learning pipeline eliminates the artificial capacity penalties imposed by worst-case deterministic modeling, unlocking the true operational potential of the distribution network.

IV. Conclusion

This paper presented an enhanced dynamic LPF optimization framework combining BESS storage and ML day-ahead solar forecasting.

Tested on a 26-bus low-voltage network in Da Nang, the model resolved end-of-line feeder bottlenecks and expanded PV hosting capacity by 72.0%, demonstrating the value of predictive intelligence for distribution grid management.

Future work could explore the integration of real-time weather data updates, the application of reinforcement learning for adaptive control strategies, and the extension of the framework to medium-voltage distribution networks.

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